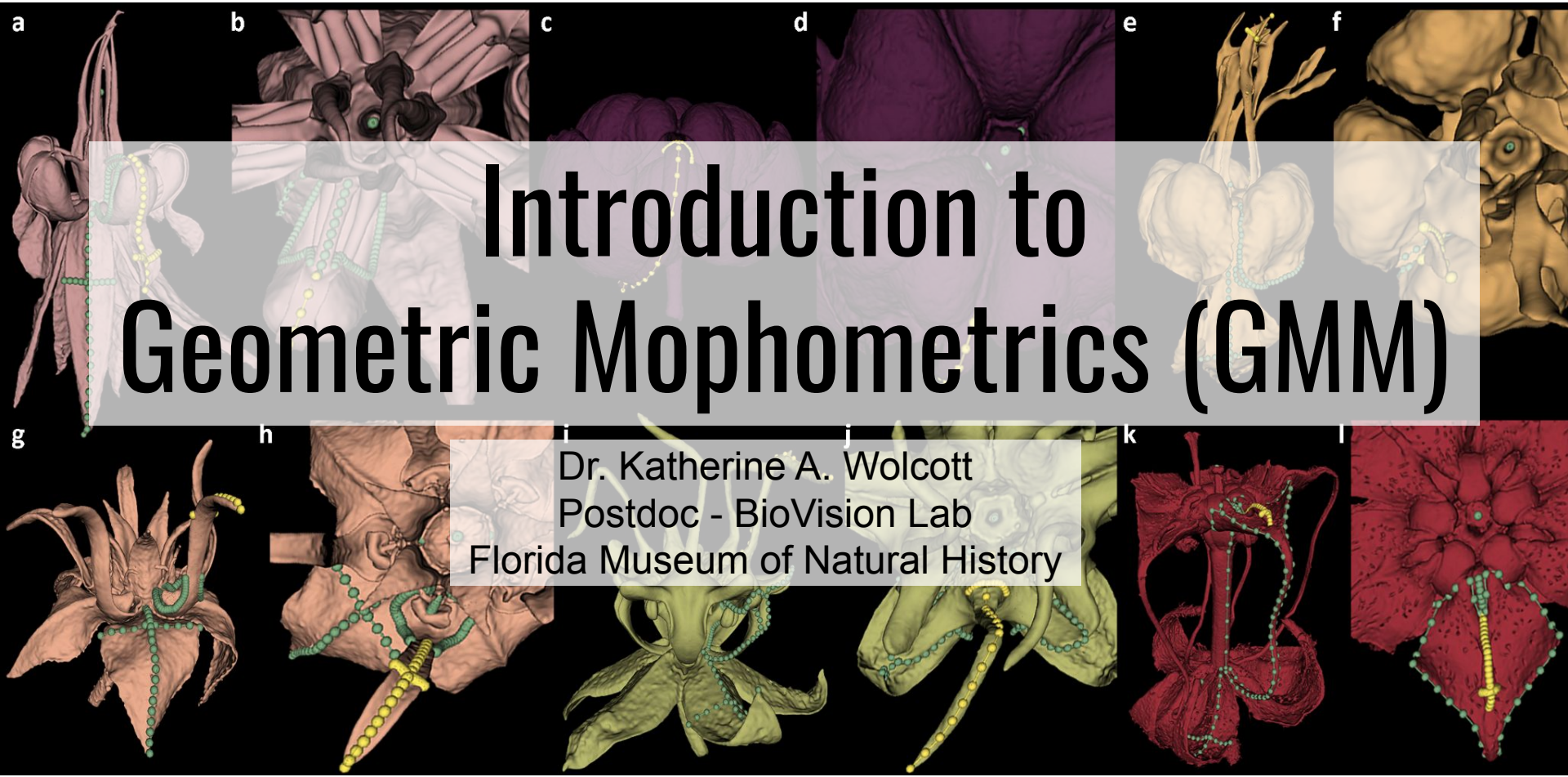
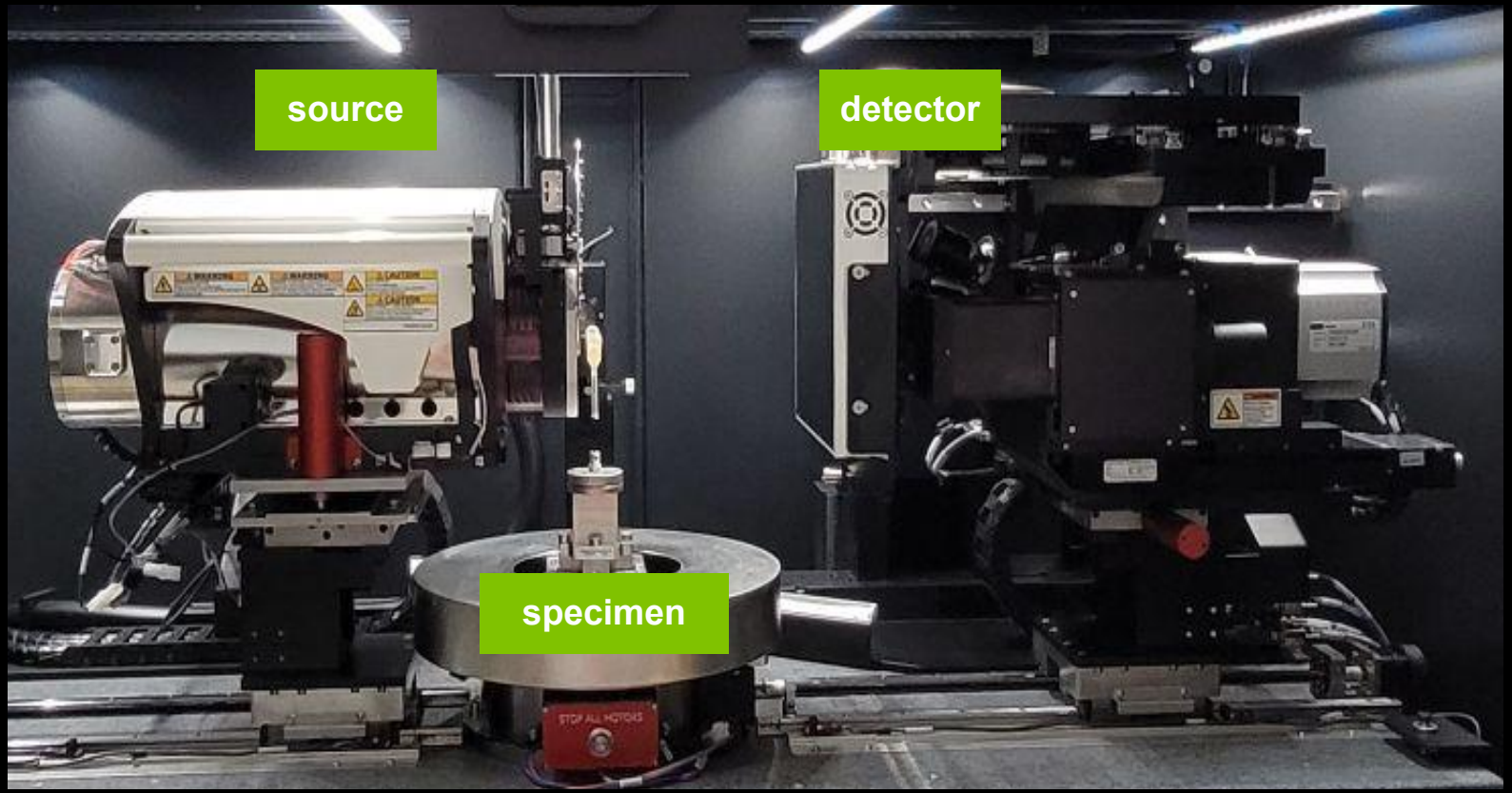


Introduction to Geometric Mophometrics (GMM)

Dr. Katherine A. Wolcott
Postdoc - BioVision Lab
Florida Museum of Natural History

Introduction to micro-CT



From specimen to micro-CT data



contrast-enhanced fixed specimen



mirco-CT scan



3D volume of micro-CT dataset

From micro-CT data to mesh



micro-CT volume



segmentation of parts



production of surfaces

Raw dataset
8-40GB
Requires special software

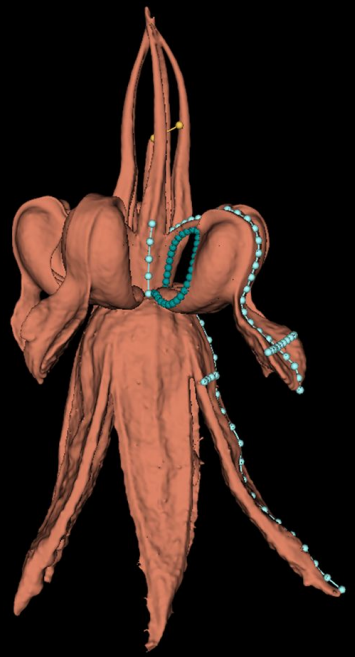
Intermediate annotated images

Processed dataset
5-50MB
Can open on any computer

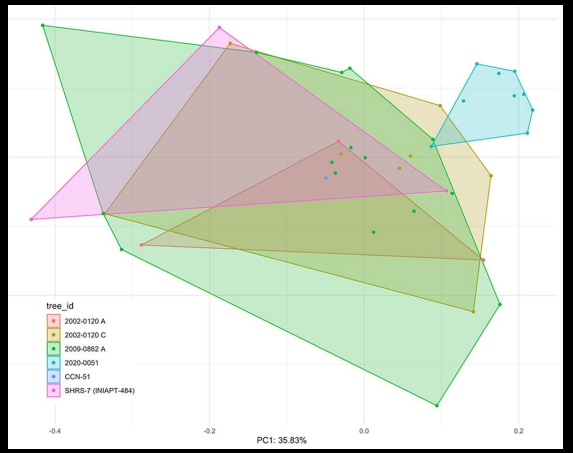
From mesh to GMM analysis



Surface Model



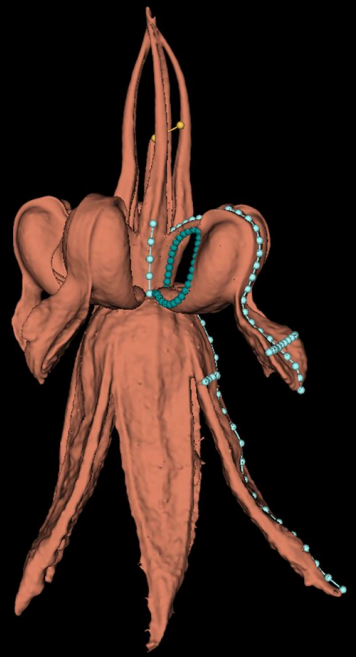
Landmarks



GMM Analysis

From mesh to GMM analysis

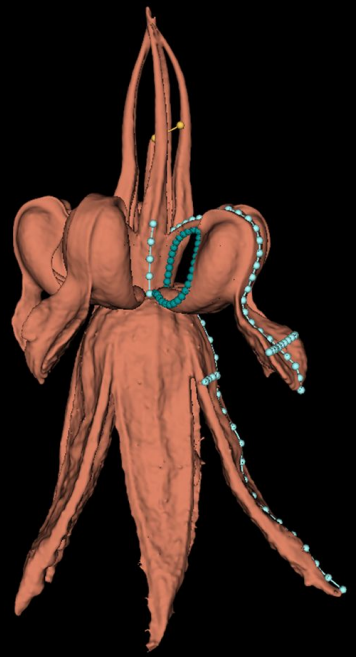




Landmarks

From mesh to GMM analysis





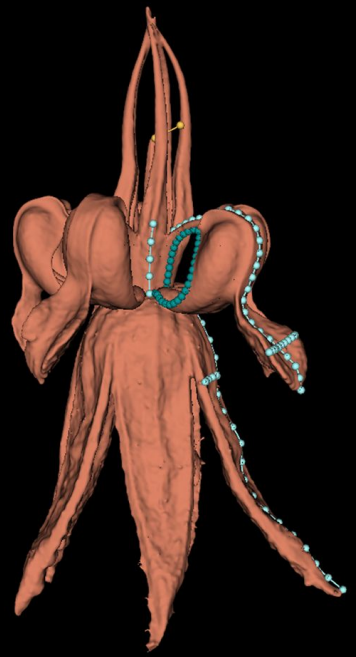
Landmarks

- Homologous
- Capture desired trait variation

From mesh to GMM analysis

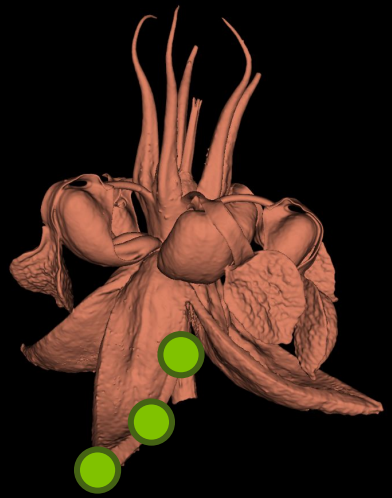


Fixed

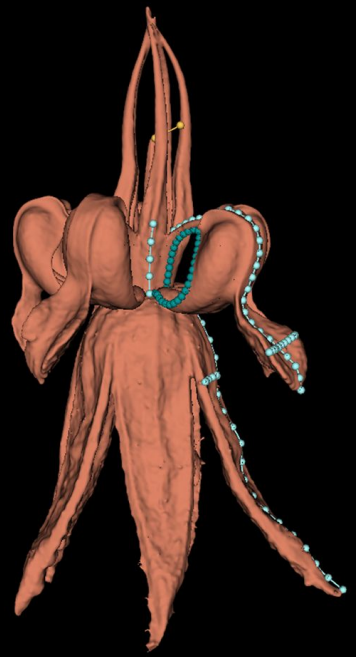


Landmarks

From mesh to GMM analysis

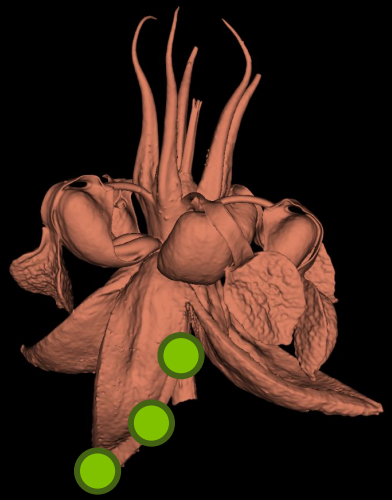


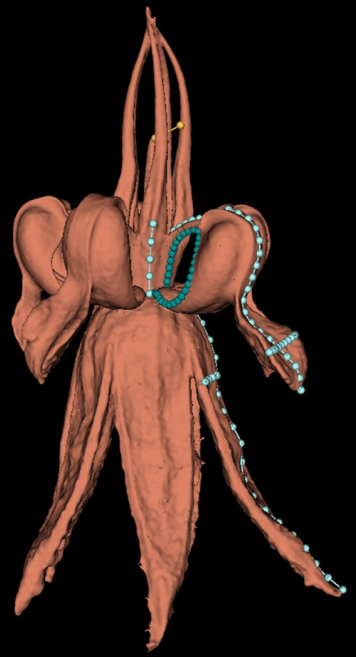
Sliding Semilandmarks



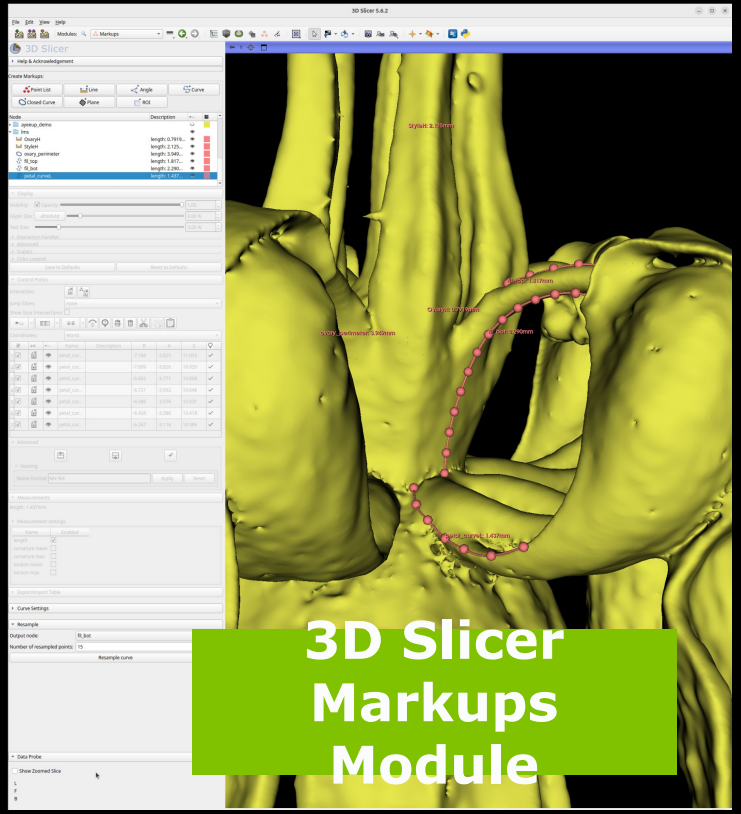
Landmarks

From mesh to GMM analysis





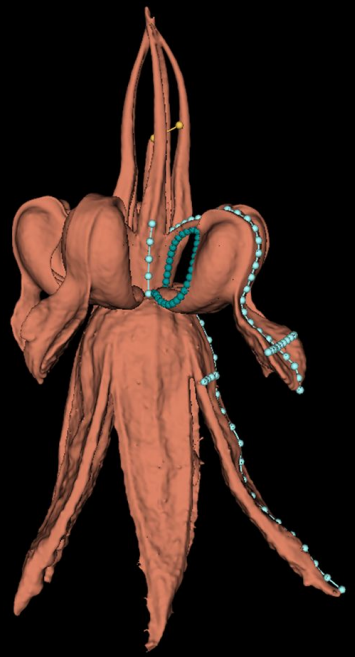
Landmarks



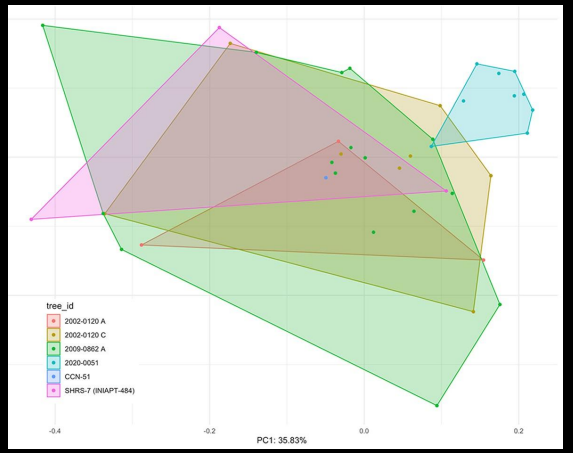
From mesh to GMM analysis



Surface Model



Landmarks

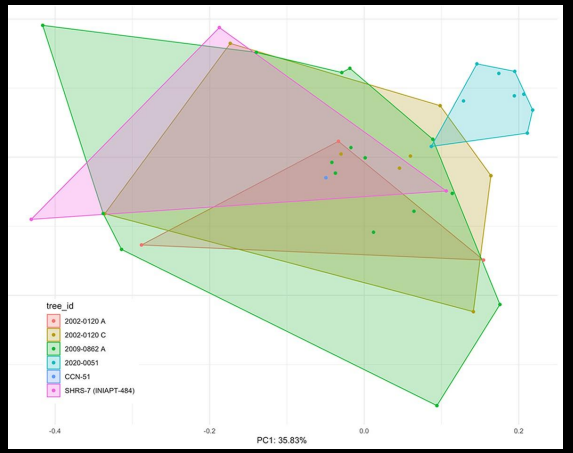


GMM Analysis

From mesh to GMM analysis

Ask questions about shape to test hypotheses

Ex: Does floral shape differ for plants growing at elevation?



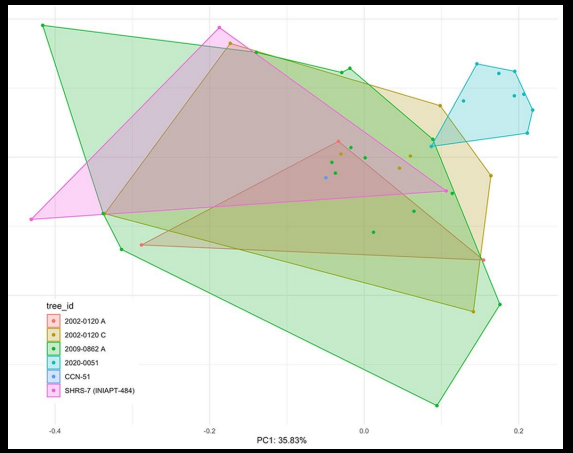
GMM Analysis

From mesh to GMM analysis

1) Landmarking + GPA
capture and align/scale flowers to focus on shape differences

2) PCA
inspect morphospace (shape change trends, overlap)

3) Linear modelling
test group effects (plant ID, elevation, locality)



GMM Analysis

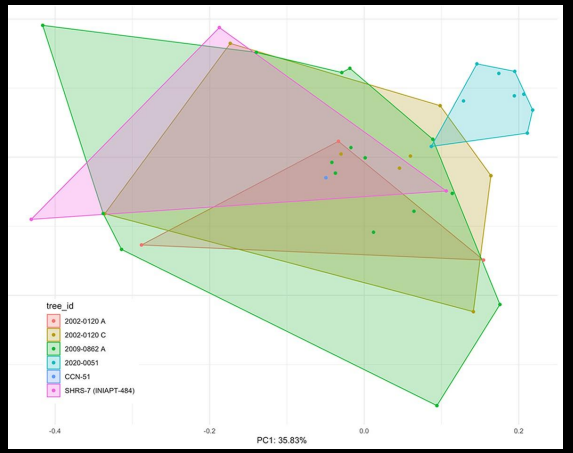
From mesh to GMM analysis

GEOMORPH

Adams et al. 2025



Rolfe et al. 2021



GMM Analysis

GMM use case: 3D pollination biology of cacao

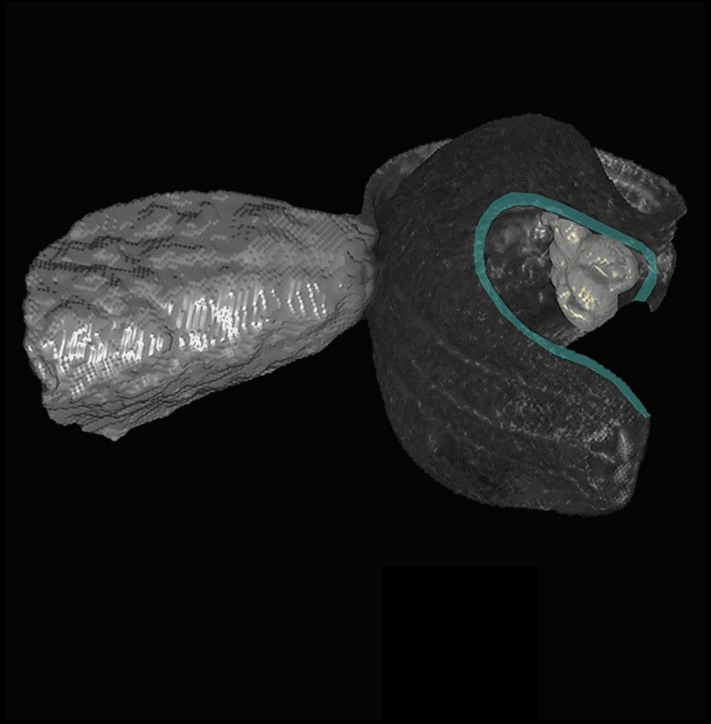


Background

- Cacao is one of the world's most economically important crops ($65 billion US value in 2020)
- Fruit set is pollinator limited
- Pollinators are unknown
  - Some suspects with limited evidence
    - Low visitation rates
    - Low pollen loads

Chumacero de Schawe et al. 2018; FAO 2023

GMM use case: 3D pollination biology of cacao



Background

- Small flowers with weird 3D morphology
  - Concealed anthers
  - Separated stigmas and anthers

- Effective pollinator needs to be able to receive and deposit pollen

- New approach to find suspected pollinators

GMM use case: 3D pollination biology of cacao



Objectives

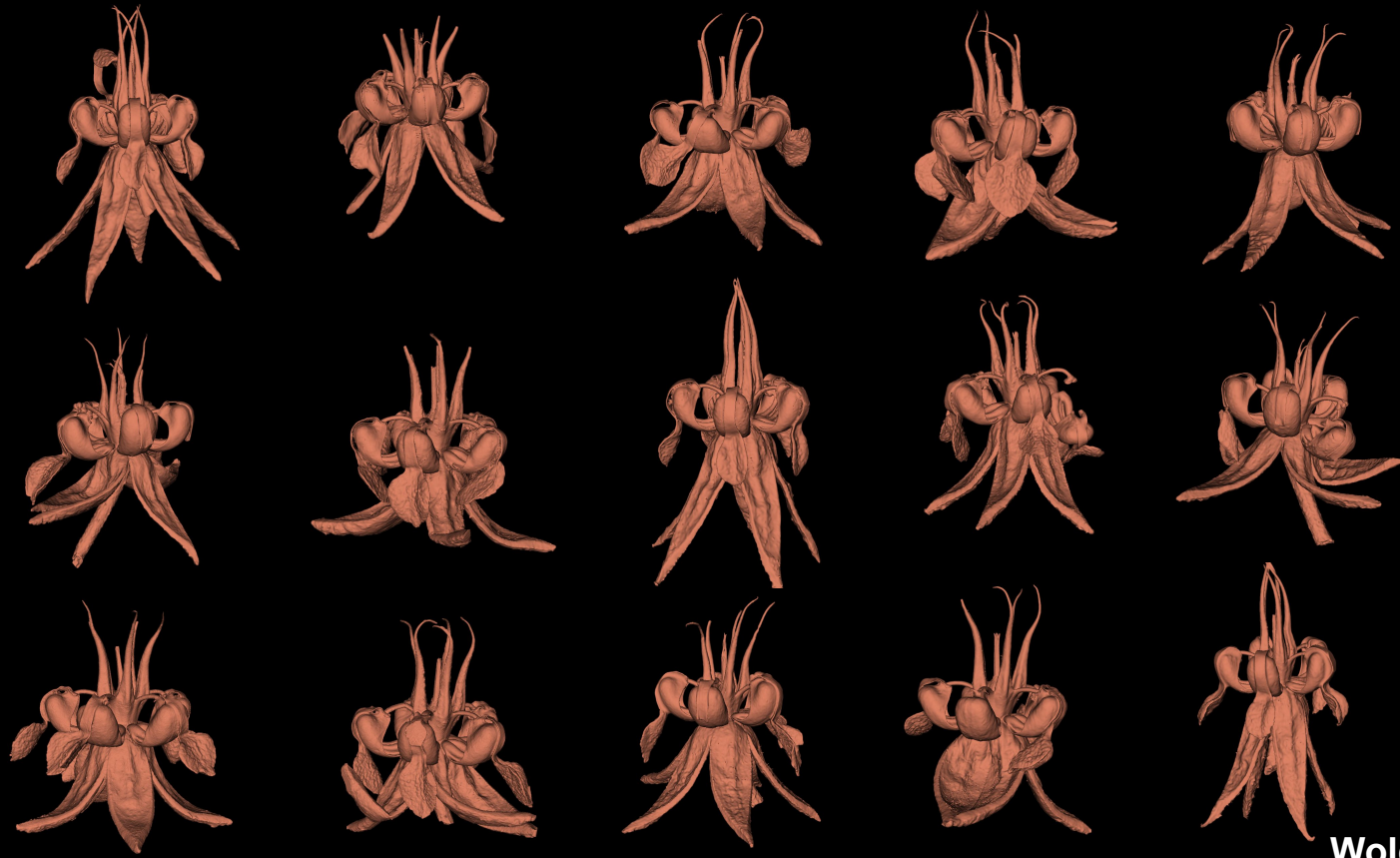
1. Obtain 50 micro-CT datasets of cacao and 20 of its hypothesized pollinator taxa

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

3. Apply method to cacao and identify pollinators that fit and do not fit within the determined size limits

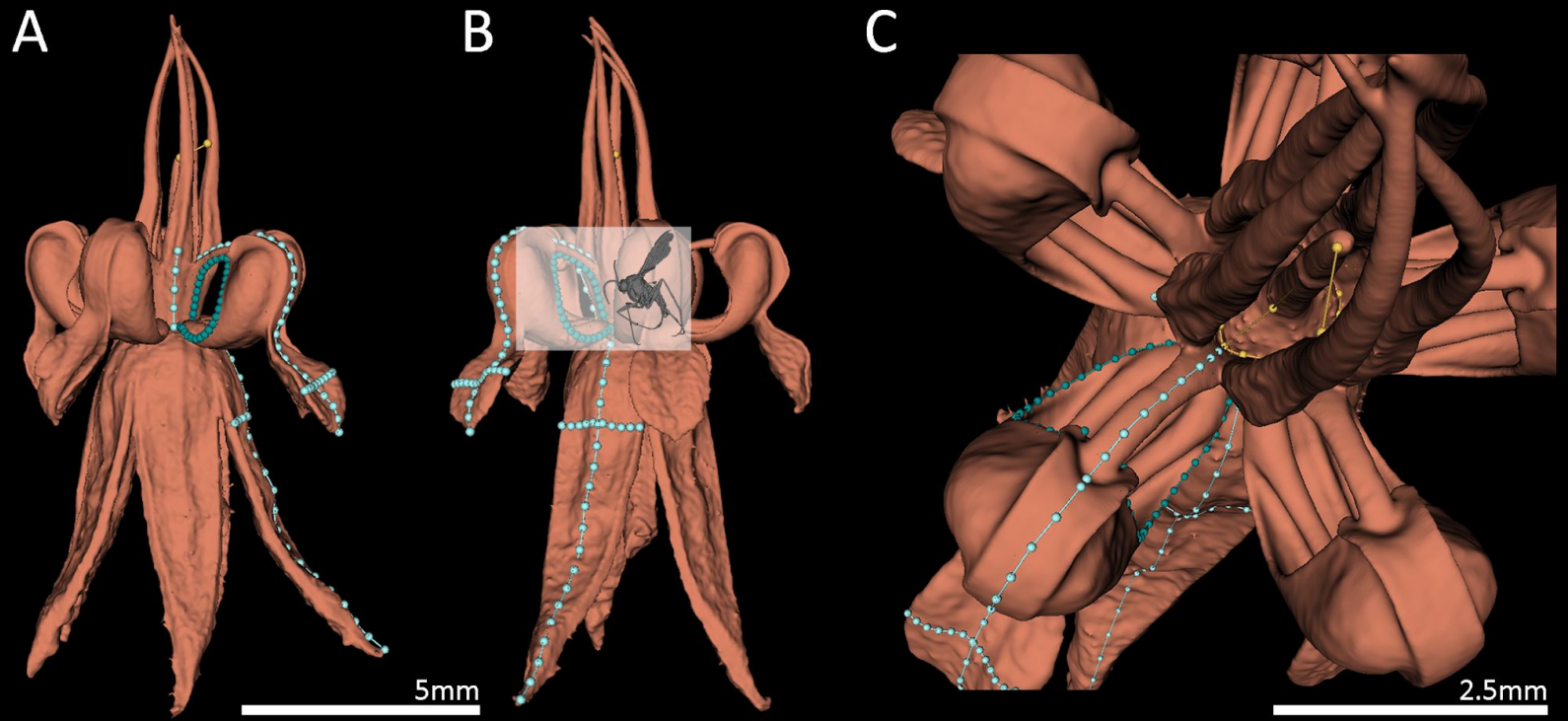
Wolcott et al. 2023

1. Obtain 50 micro-CT datasets of cacao and 20 of its hypothesized pollinator taxa



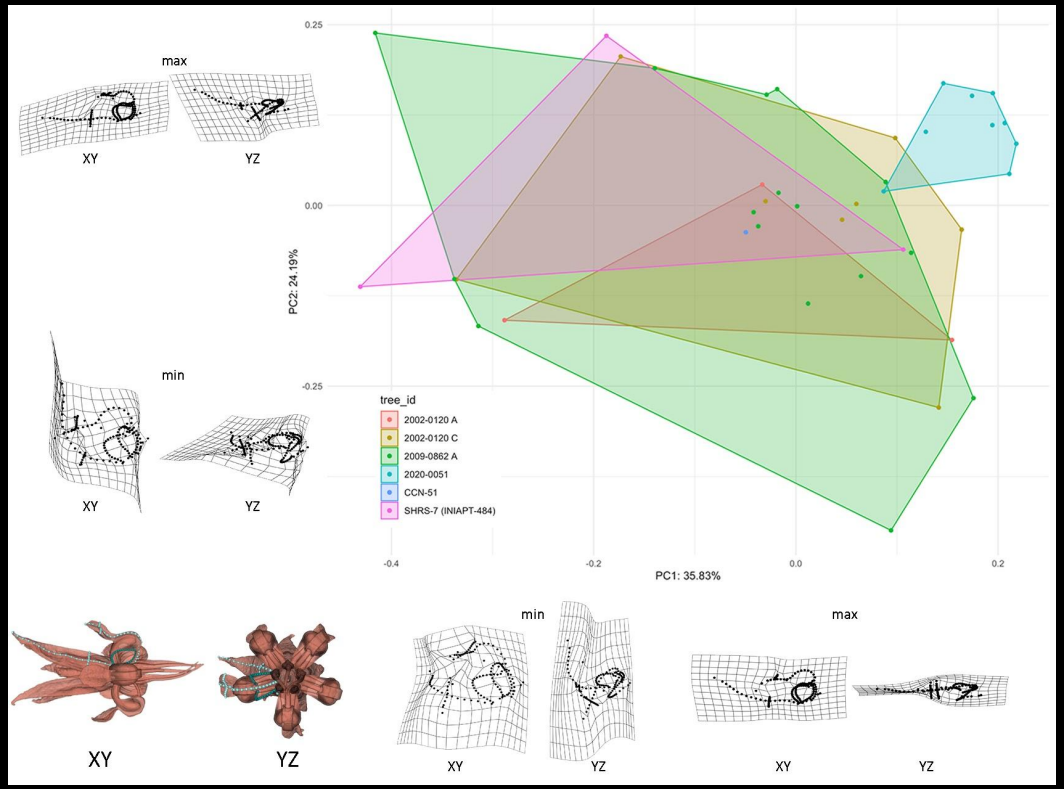
Wolcott et al. 2023

2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator



Wolcott et al. 2023

PCA: inspect morphospace



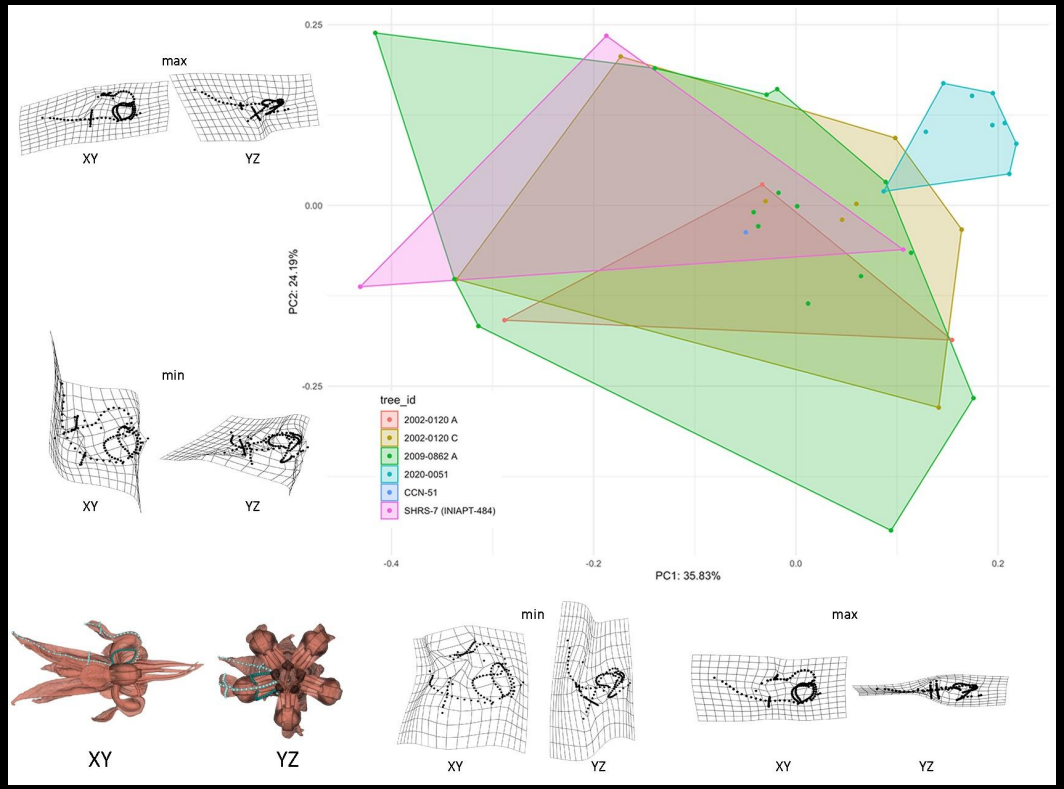
1) Are trees similar enough to treat as one group?

If yes, ask questions for all samples

If no, treat each tree separately

Wolcott et al. 2023

PCA: inspect morphospace



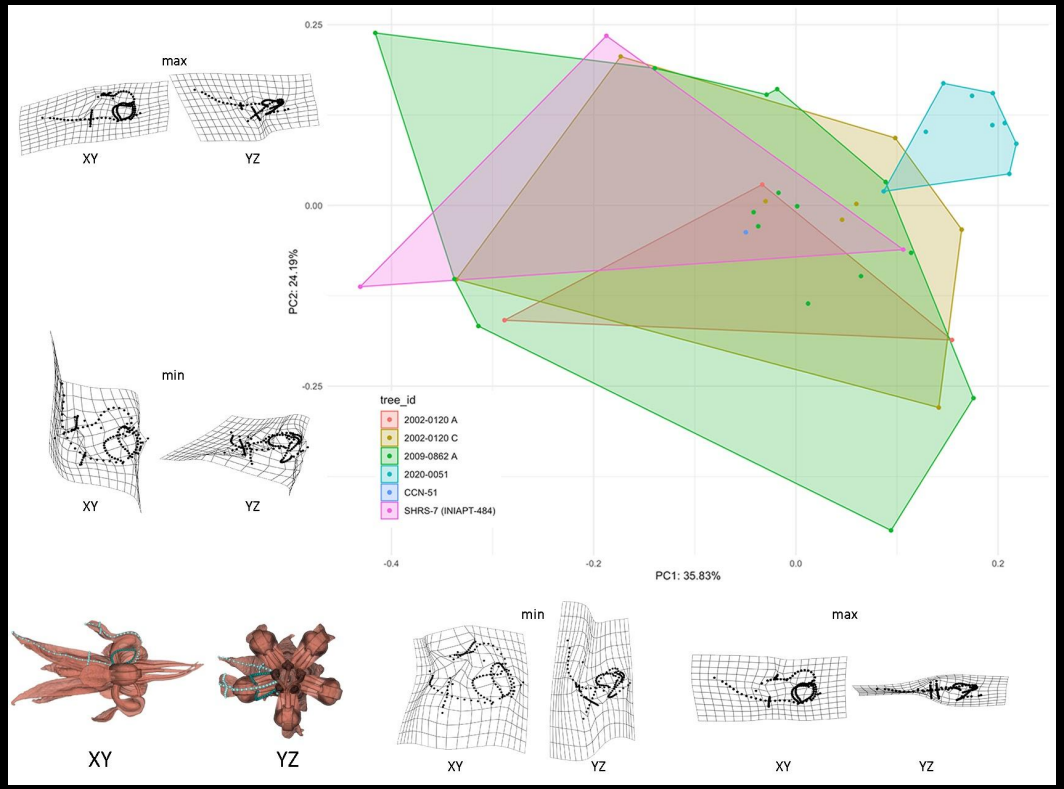
1) Are trees similar enough to treat as one group?

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If no, treat each tree separately

Wolcott et al. 2023

PCA: inspect morphospace



Principal component 1 (PC1) represents 35.83% of the variation between the floral structures, while PC2 represents 24.19% of the variation.

The flowers from all trees overlapped in the morphospace, with trees 2002-120A, 2002-120C, 2009-0862A, CCN-51, and SHRS-7 being the most similar.

Flowers from tree 2009-0862 from FTG had the greatest range of variation (shown as a green polygon), while tree 2020-0051 from FTG (shown as a blue polygon) had the least range of variation among its own flowers, but the greatest difference relative to the other trees (note the minimal overlap with other tree polygons).

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

2) Can size alone explain all differences in shape? (allometry)

Null model

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
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Total	48				

ALL

2) Can size alone explain all differences in shape? (allometry)

Null model

See example values

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	**0.900**	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL *

2) Can size alone explain all differences in shape? (allometry)

Rsq — R squared

Proportion of total shape variation a term explains

Term SS ÷ Total SS

Ranges 0 to 1 (0.125 = 12.5%)

All rows, including Residuals, sum to 1

Residuals = the share nothing in the model explains

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	**0.900**	423.00	0.001	***
Residuals	47	0.100			
Total	48				

90 %

Rsq — R squared

Proportion of total shape variation a term explains

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Ranges 0 to 1 (0.125 = 12.5%)

All rows, including Residuals, sum to 1

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Wolcott et al. 2023

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log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

2) Can size alone explain all differences in shape? (allometry)

Df — Degrees of freedom

of independent pieces of information a term uses

Continuous predictor (size) = 1

Factor with k groups = k − 1

Total = sample size − 1

Residuals = what's left over to estimate error

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

2) Can size alone explain all differences in shape? (allometry)

F — F value

Signal to noise, adjusted for Df

(Term SS ÷ Term Df) ÷ (Residual SS ÷ Residual Df)

Larger F: stronger effect relative to error

F ≈ 1 : term explains no more than noise

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

2) Can size alone explain all differences in shape? (allometry)

P — P value

How often chance produces F

Computed by permutation (eg, n=1000)

Shuffle residuals 1000× : How often does chance beat F?

P = 0.001 : essentially never

Smaller P: less likely the pattern is chance

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.900	423.00	0.001	***
Residuals	47	0.100			
Total	48				

ALL

	Df	Rsq	F	P
log(Csize)	1	0.022	1.06	0.472
Residuals	47	0.978		
Total	48			

NONE

2) Can size alone explain all differences in shape? (allometry)

See example values

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.125	6.72	0.001	***
Residuals	47	0.875			
Total	48				

2) Can size alone explain all differences in shape? (allometry)

See actual values

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	0.125	6.72	0.001	***
Residuals	47	0.875			
Total	48				

SOME

2) Can size alone explain all differences in shape? (allometry)

See actual values

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize))

	Df	Rsq	F	P	
log(Csize)	1	**0.125**	6.72	0.001	***
Residuals	47	0.875			
Total	48				

12.5 %

2) Can size alone explain all differences in shape? (allometry)

See actual values

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize) + self_compat + tree)

	Df	Rsq	F	P	
Csize	1	0.110	7.10	0.001	***
self_compat	1	0.086	5.55	0.001	***
tree	3	0.138	2.97	0.011	*
Residuals	43	0.666			
Total	48				

3) Does shape differ between groups once size is accounted for? (allometry + group effects)

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize) + self_compat + tree)

	Df	Rsq	F	P	
Csize	1	0.110	7.10	0.001	***
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tree	3	0.138	2.97	0.011	*
Residuals	43	0.666			
Total	48				

3) Does shape differ between groups once size is accounted for? (allometry + group effects)

Groups
Self-compatibility (yes or no)
Tree (5 trees)

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize) + self_compat + tree)

	Df	Rsq	F	P	
Csize	1	0.110	7.10	0.001	***
self_compat	1	0.086	5.55	0.001	***
tree	3	0.138	2.97	0.011	*
Residuals	43	0.666			
Total	48				

Yes (?)

3) Does shape differ between groups once size is accounted for? (allometry + group effects)

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Wolcott et al. 2023

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tree	3	0.138	2.97	0.011	*
Residuals	43	0.666			
Total	48				

Yes (?)

Self-compatibility is measured per tree, Cannot separate effects at this sample size

3) Does shape differ between groups once size is accounted for? (allometry + group effects)

Groups
Self-compatibility (yes or no)
Tree (5 trees)

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize) * self_compat * tree)

	Df	Rsq	F	P
log(Csize)	1	0.110	7.20	0.001 ***
self_compat	1	0.086	5.63	0.001 ***
tree	3	0.138	3.01	0.009 **
log(Csize):self_compat	1	0.020	1.31	0.248
log(Csize):tree	3	0.050	1.09	0.371
Residuals	39	0.596		
Total	48			

4) Do groups differ in how shape changes with size?
(allometry * group effects)

Groups
Self-compatibility (yes or no)
Tree (5 trees)

Wolcott et al. 2023

Linear modelling: test group effects

procD.lm(f1 = coords ~ log(Csize) * self_compat * tree)

	Df	Rsq	F	P
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self_compat	1	0.086	5.63	0.001 ***
tree	3	0.138	3.01	0.009 **
log(Csize):self_compat	1	0.020	1.31	0.248
log(Csize):tree	3	0.050	1.09	0.371
Residuals	39	0.596		
Total	48			

NO

4) Do groups differ in how shape changes with size?
(allometry * group effects)

Groups
Self-compatibility (yes or no)
Tree (5 trees)

Wolcott et al. 2023

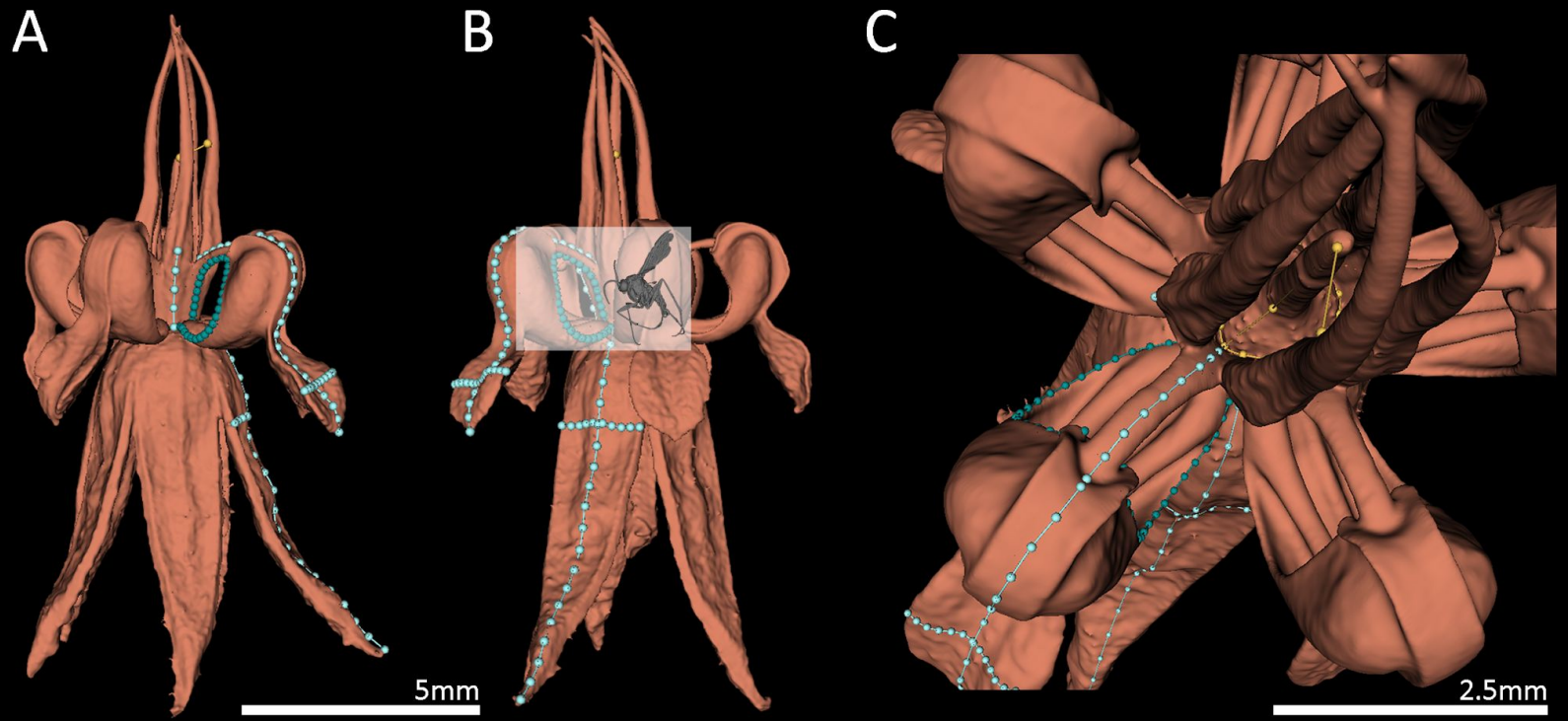
2. Develop a method using geometric morphometrics (GMM) to determine the functional size limits of a pollinator

Shape varies by size, tree, and self-compatibility

Through our one-way Procrustes ANOVA, flower size ($F_{1,39} = 5.686$, $P = 0.001$), tree self-compatibility ($F_{1,39} = 4.463$, $P = 0.001$), and tree individual ($F_{4,39} = 1.790$, $P = 0.045$) were statistically significant predictors of floral shape, while collection month was not ($F_{2,39} = 1.774$, $P = 0.071$; Appendix S6).

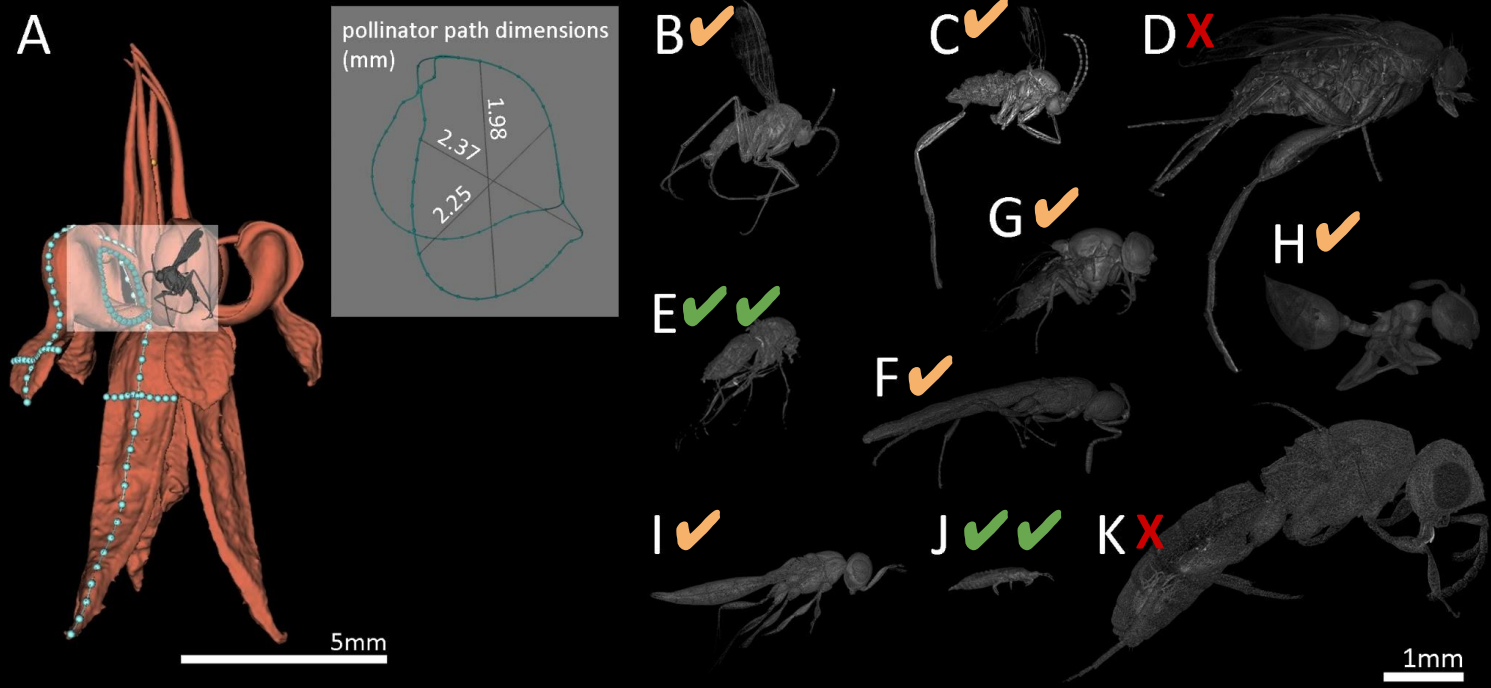
Wolcott et al. 2023

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Wolcott et al. 2023

3. Apply method to cacao and identify pollinators that fit and do not fit within the determined size limits



Wolcott et al. 2023

GMM use case: 3D pollination biology of cacao



Objectives

1. Obtain 50 micro-CT datasets of cacao and 20 of its hypothesized pollinator taxa

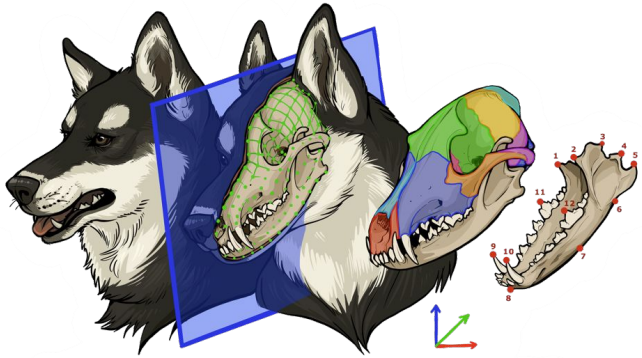
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Wolcott et al. 2023

How to do your own 3D GMM analysis?

GEOMORPH

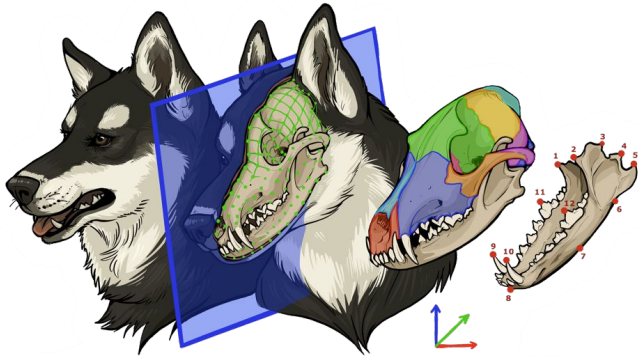


Adams et al. 2025

Rolfe et al. 2021

How to do your own 3D GMM analysis?

GEOMORPH



Workshop Materials on GitHub

SlicerMorph Tutorial
Demo Data - Lab Mice Skulls, LMs, (fake) metadata
Google Colab Geomorph demo



Questions?

References

Adams et al. 2013. Geomorph: An r package for the collection and analysis of geometric morphometric shape data. MEE. https://doi.org/10.1111/2041-210X.12035

Chumacero de Schawe et al. 2018. Abundance and diversity of flower visitors on wild and cultivated cacao (Theobroma cacao L.) in Bolivia. Agr. Sys.

Food and Agriculture Organization of the United Nations. 2023. FAOSTAT statistical database. [accessed 3 February 2023].

Rolfe et al. 2021. SlicerMorph: An open and extensible platform to retrieve, visualize and analyse 3D morphology. MEE. https://doi.org/10.1111/2041-210X.13669

Wolcott et al. 2023. 3D pollination biology using micro - CT and geometric morphometrics in *Theobroma cacao* L. APPS. https://doi.org/10.1002/aps3.11549